

Avinash Parmar

📞 7033980391 — ✉ akavinashparmar@gmail.com — 📄 Avinash Parmar — 🔄 akavinashparmar2006 — 📁 LeetCode — 🌐 Portfolio

PROFILE SUMMARY

B.Tech Artificial Intelligence and Data Science student with a strong foundation in software engineering, full-stack development, data science, and machine learning. Proficient in Python, Java, JavaScript, HTML, CSS, React.js, Node.js, MySQL, and MongoDB, with hands-on experience developing REST APIs, database-driven applications, and machine learning solutions. Strong understanding of Data Structures and Algorithms, Object-Oriented Programming, DBMS, SQL, and software development principles. Experienced in data preprocessing, machine learning model deployment, API integration, and building practical technology solutions. Strong problem-solving, analytical, teamwork, and leadership abilities with an interest in Software Engineering, Full-Stack Development, and Machine Learning roles.

EDUCATION

Bachelor of Technology (Artificial Intelligence and Data Science) CGC University, Mohali	2024-2028 CGPA: 8.34
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Senior Secondary School Cambridge Senior Secondary	2023
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Secondary School Cambridge Senior Secondary	2021
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PROJECTS

AI-Based Health Prediction	Mar 2026 – April 2026
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- Developed a full-stack health monitoring application using Java, MySQL, REST APIs, and OpenAI API for health data management and AI-assisted insights.
- Implemented daily health logging for sleep, steps, water intake, calories, and other health metrics, storing historical records in a MySQL database.
- Developed an AI-assisted prediction system that analyzes historical user data and generates personalized health insights and recommendations.
- Designed role-based User and Admin portals for health tracking, user management, data administration, and dashboard-based monitoring.
- Implemented health score calculation and historical trend analysis using stored user health records and SQL-based data management.

Technologies: Java, MySQL, SQL, OpenAI API, REST APIs

TerraMetric – Land Management & GIS Platform

- Built a full-stack GIS-based land management platform using Spring Boot, MySQL, FastAPI, and JavaScript for scalable application development.
- Developed machine learning microservices for NDVI analysis, land classification, and field segmentation to support data-driven land management.
- Implemented secure authentication and authorization using JWT, OAuth2, and role-based access control (RBAC).
- Developed REST APIs and backend services to integrate the frontend, relational database, and machine learning microservices.
- Enabled real-time updates, geospatial tools, and blockchain-based tracking to improve transparency and land transaction management.

Technologies: Spring Boot, MySQL, FastAPI, JavaScript, REST APIs, JWT, OAuth2

California Housing Value Estimator	Recent
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- Developed a machine learning-based housing value prediction web application using the California Housing dataset with 20,000+ records and 9 key features.
- Built a Flask REST API integrating a pre-trained Scikit-learn model and preprocessing pipeline for real-time housing value prediction without retraining during inference.
- Utilized Pandas, NumPy, and Scikit-learn for data preprocessing, feature processing, machine learning workflows, and model inference.

- Used Joblib to serialize and load the trained machine learning model and preprocessing artifacts for efficient production inference.
- Deployed the application on Render, providing a production-ready web interface and REST API for model predictions.

Technologies: Python, Flask, Pandas, NumPy, Scikit-learn, Joblib, HTML/CSS, REST API, Machine Learning, Render

SKILLS

Programming Languages: Python, Java, JavaScript, SQL

Web Technologies: HTML, CSS, React.js (Basics), Node.js (Basics), REST APIs, Full-Stack Development

Data Science & ML: NumPy, Pandas, Matplotlib, Scikit-learn (Sklearn), Data Preprocessing, ML Pipelines, Model Inference, Machine Learning Deployment

Core Computer Science: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Software Engineering, Full-Stack Development

Databases & APIs: MySQL, MongoDB, SQL, REST APIs, API Integration, CRUD Operations

Tools & Technologies: Git, GitHub, Flask, FastAPI, Spring Boot, Joblib, Jupyter Notebook

Soft Skills: Leadership, Flexibility, Active Listening, Critical Thinking, Problem Solving, Teamwork, Communication

ACHIEVEMENTS / EXTRA-CURRICULAR

– Technical Lead – Student Community

Founded and led a student technical community, guiding peers in project development, software architecture, coding practices, and technical problem solving.

– Hackathons & Innovation

Participated in multiple hackathons, strengthening rapid problem-solving, teamwork, technical collaboration, and development of practical technology solutions under time constraints.

– Competitive Programming

Solved problems across platforms including LeetCode, GeeksforGeeks, and HackerRank, strengthening Data Structures and Algorithms, problem-solving, and algorithmic thinking skills.

CERTIFICATIONS

– Ultimate Web Development – Code With Harry

Gained practical knowledge of front-end and back-end web development, JavaScript, HTML, CSS, and full-stack development fundamentals.

– Data Scientist – Code With Harry

Covered data science concepts, Python-based data analysis, data preprocessing, machine learning fundamentals, and analytical techniques.

– Programming Fundamentals Using Python – Infosys

Developed strong foundations in Python programming, data structures, algorithms, programming logic, and problem-solving techniques.